

THE TENTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 25, 1950

Morning Session

1. For what values of the ratio a/b is the limaçon $r = a - b \cos \theta$ a convex curve? ($a > b > 0$) (page 288)

2. Answer both (i) and (ii). Test for convergence the series

$$(i) \frac{1}{\log(2!)} + \frac{1}{\log(3!)} + \frac{1}{\log(4!)} + \cdots + \frac{1}{\log(n!)} + \cdots$$

$$(ii) \frac{1}{3} + \frac{1}{3\sqrt{3}} + \frac{1}{3\sqrt{3}\sqrt[3]{3}} + \cdots + \frac{1}{3\sqrt{3}\sqrt[3]{3} \cdots \sqrt[n]{3}} + \cdots$$

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3. The sequence $x_0, x_1, x_2, \dots$ is defined by the conditions

$$x_0 = a, x_1 = b, x_{n+1} = \frac{x_{n-1} + (2n-1)x_n}{2n} \quad \text{for } n \geq 1,$$

where a and b are given numbers. Express $\lim_{n \rightarrow \infty} x_n$ concisely in terms of a and b . (page 291)

4. Answer either (i) or (ii).

(i) In a right prism with triangular base, given the sum of the areas of three mutually adjacent faces (that is, of two lateral faces and one base), show that these faces are of equal area and perpendicular to each other when the volume attains its maximum. (page 292)

(ii) Show that

$$\frac{\frac{x}{1} + \frac{x^3}{1 \cdot 3} + \frac{x^5}{1 \cdot 3 \cdot 5} + \frac{x^7}{1 \cdot 3 \cdot 5 \cdot 7} + \cdots}{1 + \frac{x^2}{2} + \frac{x^4}{2 \cdot 4} + \frac{x^6}{2 \cdot 4 \cdot 6} + \cdots} = \int_0^x e^{-t^2/2} dt.$$

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5. A function $D(n)$ of the positive integral variable n is defined by the

following properties: $D(1) = 0$, $D(p) = 1$ if p is a prime, $D(uv) = uD(v) + vD(u)$ for any two positive integers u and v . Answer all three parts below.

(i) Show that these properties are compatible and determine uniquely $D(n)$. (Derive a formula for $D(n)/n$, assuming that $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ where $p_1, p_2, \dots, p_k$ are different primes.)

(ii) For what values of n is $D(n) = n$?

(iii) Define $D^2(n) = D[D(n)]$, etc., and find the limit of $D^m(63)$ as m tends to ∞ . (page 294)

6. Each coefficient a_n of the power series

$$a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots = f(x)$$

has either the value 1 or the value 0. Prove the easier of the two assertions:

(i) If $f(0.5)$ is a rational number, $f(x)$ is a rational function.

(ii) If $f(0.5)$ is not a rational number, $f(x)$ is not a rational function.

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Afternoon Session

1. In each of n houses on a straight street are one or more boys. At what point should all the boys meet so that the sum of the distances that they walk is as small as possible? (page 299)

2. Two obvious approximations to the length of the perimeter of the ellipse with semi-axes a and b are $\pi(a + b)$ and $2\pi(ab)^{1/2}$. Which one comes nearer the truth when the ratio b/a is very close to 1? (page 300)

3. In the Gregorian calendar:

- (i) years not divisible by 4 are common years;
- (ii) years divisible by 4 but not by 100 are leap years;
- (iii) years divisible by 100 but not by 400 are common years;
- (iv) years divisible by 400 are leap years;
- (v) a leap year contains 366 days; a common year 365 days.

Prove that the probability that Christmas falls on a Wednesday is not $1/7$. (page 302)

4. The cross-section of a right cylinder is an ellipse, with semi-axes a and b , where $a > b$. The cylinder is very long, made of very light homogeneous material. The cylinder rests on the horizontal ground which it touches along the straight line joining the lower endpoints of the minor axes of its several cross-sections. Along the upper endpoints of these minor axes lies a very heavy homogeneous wire, straight and just as long as the cylinder. The wire

and the cylinder are rigidly connected. We neglect the weight of the cylinder, the breadth of the wire, and the friction of the ground.

The system described is in equilibrium, because of its symmetry. This equilibrium seems to be stable when the ratio b/a is very small, but unstable when this ratio comes close to 1. Examine this assertion and find the value of the ratio b/a which separates the cases of stable and unstable equilibrium.

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5. Answer either (i) or (ii).

(i) Given that the sequence whose n th term is $(s_n + 2s_{n+1})$ converges, show that the sequence $\{s_n\}$ converges also.

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(ii) A plane varies so that it includes a cone of constant volume equal to $\pi a^3/3$ with the surface the equation of which in rectangular coordinates is $2xy = z^2$. Find the equation of the envelope of the various positions of this plane.

State the result so that it applies to a general cone (that is, conic surface) of the second order.

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6. Consider the closed plane curves C_i and C_o , their respective lengths $|C_i|$ and $|C_o|$, the closed surfaces S_i and S_o , and their respective areas $|S_i|$ and $|S_o|$. Assume that C_i lies inside C_o and S_i inside S_o . (Subscript i stands for "inner," o for "outer.") Prove the correct assertions among the following four, and disprove the others.

(i) If C_i is convex, $|C_i| \leq |C_o|$.

(ii) If S_i is convex, $|S_i| \leq |S_o|$.

(iii) If C_o is the smallest convex curve containing C_i , then $|C_o| \leq |C_i|$.

(iv) If S_o is the smallest convex surface containing S_i , then $|S_o| \leq |S_i|$.

You may assume that C_i and C_o are polygons and S_i and S_o polyhedra. (Why?)

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